



DAFFODIL INTERNATIONAL UNIVERSITY

Department of Computing and Information System (CIS)

Web Development Essentials Lab

LAB Final Project

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Project Report

1. Project Title

LearnC — Learning C Programming Website

LearnC is a simple educational website for beginners who want to learn the basics of C programming.

2. Project Overview

LearnC website provides basic C programming topics, simple C programs, and a quiz. The website is designed with a simple and user-friendly interface so that beginners can easily learn C programming.

The project demonstrates practical understanding of:

- Semantic HTML structure
 - CSS styling
 - Responsive web design
 - Bootstrap components
 - JavaScript basics
 - JavaScript functions
 - JavaScript objects
 - DOM manipulation
 - Event handling
 - Basic conditional statements
 - Interactive webpage elements
-

3. Objectives

The main objectives of this project are:

- To create a simple C programming learning website.
- To provide basic C programming concepts.
- To show simple C programming examples.
- To practice HTML and CSS.
- To implement responsive web design.
- To use JavaScript for interactive features.
- To understand DOM manipulation and event handling.
- To create a user-friendly website for beginners.

4. Project Outline / Site Structure

The website contains the following main sections:

Home

The Home section introduces the LearnC website and provides buttons to start learning and view programs.

Topics

The Topics section contains eight basic C programming topics:

1. Introduction to C
2. Variables
3. Data Types
4. Operators
5. If / Else
6. Loops
7. Functions
8. Arrays

Each topic provides a short explanation, key points, and a simple code example.

Programs

The Programs section contains six beginner-friendly C programs:

1. Hello World
2. Addition
3. Even / Odd
4. Largest Number
5. Factorial
6. Multiplication Table

Each program shows the C code and expected output.

Login / Sign Up

The website includes a simple Login and Sign Up system. User information is stored in the browser using local storage.

5. Tools and Technologies Used

The following technologies are used:

Technology	Use
HTML5	Website structure
CSS3	Website design
Bootstrap	Responsive design
JavaScript	Interactive features
Bootstrap Icons	Icons

6. Key Features

The main features of the LearnC website are:

- Responsive navigation bar
- Topics dropdown menu
- Programs dropdown menu
- Eight C programming topics
- Six beginner C programs
- Interactive topic modal
- Interactive program modal
- Login system
- Sign Up system
- Local Storage
- JavaScript DOM manipulation
- JavaScript functions
- Interactive buttons
- Responsive design
- Bootstrap Icons

The JavaScript dynamically displays topic information and program details inside modal windows.

7. Responsive Design

The website is designed to work on different screen sizes using CSS media queries.

Device	Screen Size	Main Adjustment
Desktop	Above 1050px	Full navigation and multi-column layout
Tablet	821px – 1050px	Reduced spacing and two-column cards
Small Tablet	601px – 820px	Stacked sections and adjusted navigation
Mobile	Up to 600px	Single-column layout and smaller fonts
Small Mobile	Up to 380px	Further reduced spacing and font sizes

The CSS contains media queries for 1050px, 820px, 600px and 380px screen sizes.

8. Folder Structure

The project is organized into separate files and folders.

LearnC/

|

|— index.html

|— style.css

|— script.js

Here,

The HTML files contain the structure and content of the website.

CSS Folder:

The style.css file contains the custom design and responsive styling.

JS Folder:

The script.js file contains the JavaScript functions and interactive features.

9. Challenges and Learnings

During the development of the LearnC project, several important web development concepts were practiced.

HTML

The project helped improve understanding of:

- Semantic HTML
- Page structure
- Navigation
- Forms
- Links
- Sections
- Tables
- Images

CSS

The project helped develop knowledge of:

- Flexbox
- Spacing
- Margin and padding
- Colors
- Buttons
- Cards
- Hover effects
- Media queries
- Responsive layouts

Bootstrap

Bootstrap was used to understand:

- Containers
- Grid system
- Responsive classes
- Buttons
- Cards
- Utility classes
- Bootstrap Icons

JavaScript

The project mainly helped practice beginner-level JavaScript concepts.

These include:

- Variables
- Functions
- Objects
- If/else
- DOM manipulation
- Events
- getElementById()
- classList
- Updating HTML content

A major learning point was understanding how JavaScript can interact with HTML elements and make a static webpage interactive.

10. Source Code

Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>C Learn | Learn C Programming</title>
  <link rel="icon" type="image/png" href="C.png">
  <link href="https://cdn.jsdelivr.net/npm/bootstrap-
icons@1.13.1/font/bootstrap-icons.min.css" rel="stylesheet">
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header class="navbar">
    <div class="nav-container">
      <a href="#home" class="logo" onclick="goHome()" ">
        Learn<span>C</span>
      </a>
    <nav class="nav-menu">
```

```
<a href="#home" onclick="goHome()">Home</a>

<div class="dropdown">
    <button class="nav-button"
onclick="toggleMenu('topicsMenu')">
        Topics <span>▼</span>
    </button>
    <div class="dropdown-menu" id="topicsMenu">
        <button onclick="openTopic(0)">C Introduction</button>
        <button onclick="openTopic(1)">Variables</button>
        <button onclick="openTopic(2)">Data Types</button>
        <button onclick="openTopic(3)">Operators</button>
        <button onclick="openTopic(4)">If / Else</button>
        <button onclick="openTopic(5)">Loops</button>
        <button onclick="openTopic(6)">Functions</button>
        <button onclick="openTopic(7)">Arrays</button>
    </div>
</div>

<div class="dropdown">
    <button class="nav-button"
onclick="toggleMenu('programsMenu')">
        Programs <span>▼</span>
    </button>
    <div class="dropdown-menu" id="programsMenu">
        <button onclick="openProgram(0)">Hello World</button>
        <button onclick="openProgram(1)">Addition</button>
        <button onclick="openProgram(2)">Even / Odd</button>
        <button onclick="openProgram(3)">Largest
Number</button>
        <button onclick="openProgram(4)">Factorial</button>
        <button onclick="openProgram(5)">Multiplication
Table</button>
    </div>
```

```

        </div>

        <a class="disabled-link" href="#" onclick="return false;"
title="Quiz will be added later">Quiz</a>

        <a class="disabled-link" href="#" onclick="return false;"
title="Contact will be added later">Contact</a>

    </nav>

    <div class="nav-actions">

        <button class="login-btn" onclick="openAuth('login') ">Log
In</button>

        <button class="signup-btn" onclick="openAuth('signup') ">Sign
Up</button>

    </div>

</div>

</header>

<main id="home">

    <section class="hero">

        <div class="hero-container">

            <div class="hero-content">

                <p class="eyebrow">WELCOME TO LEARN C</p>

                <h1>Learn C Programming<br><span>Simply &
Easily</span></h1>

                <p class="hero-text">

                    Learn the basics of C Programming with simple
explanations,

                    examples and beginner-friendly programs.

                </p>

                <div class="hero-buttons">

                    <button class="primary-btn"
onclick="showSection('topicsSection') ">

                        Start Learning <i class="bi bi-arrow-right"></i>

                    </button>

                    <button class="outline-btn"
onclick="showSection('programsSection') ">

```

```
View Programs

</button>

</div>

<p class="hero-note"><i class="bi bi-check-circle"></i> 8
core topics &nbsp;• &nbsp;6 beginner programs</p>

</div>

<div class="code-card">
  <div class="code-header">
    <span></span><span></span><span></span>
    <small>hello.c</small>
  </div>
  <pre><code>#include <stdio.h>

int main()
{
    printf("Hello World!");

    return 0;
}</code></pre>
</div>
</div>
</section>

<section class="intro section">
  <div class="section-heading">
    <p class="eyebrow">ABOUT C</p>
    <h2>What is C Programming?</h2>
    <p>C is a general-purpose programming language that is
excellent for learning programming fundamentals.</p>
  </div>

  <div class="intro-grid">
```

```

    <div class="intro-text">
        <h3>A Great Language for Beginners</h3>
        <p>
            C helps students understand important programming ideas
such as
            variables, conditions, loops, functions and arrays.
        </p>
        <p>
            This website keeps the lessons short and practical so
you can
            learn one concept at a time and immediately see a
simple example.
        </p>
        <button class="primary-btn"
onclick="showSection('topicsSection')">Explore Topics</button>
    </div>

    <div class="info-box">
        <div class="info-item">
            <i class="bi bi-lightbulb"></i>
            <div><h4>Easy to Understand</h4><p>Learn programming
step by step.</p></div>
        </div>
        <div class="info-item">
            <i class="bi bi-code-square"></i>
            <div><h4>Practice Programs</h4><p>Practice six simple C
programs.</p></div>
        </div>
        <div class="info-item">
            <i class="bi bi-book"></i>
            <div><h4>Strong Basics</h4><p>Build a clear foundation
in C.</p></div>
        </div>
    </div>
</div>

```

```

</section>

<section class="topics-section section" id="topicsSection">
  <div class="section-heading">
    <p class="eyebrow">8 CORE TOPICS</p>
    <h2>What Will You Learn?</h2>
    <p>Click any topic to see a short explanation, syntax and
beginner example.</p>
  </div>

  <div class="card-grid topics-grid">
    <button class="topic-card" onclick="openTopic(0)">
      <span class="card-number">01</span><i class="bi bi-code-
slash"></i>
      <h3>C Introduction</h3><p>Understand C, its structure and
your first program.</p>
      <span class="learn-link">Learn topic →</span>
    </button>
    <button class="topic-card" onclick="openTopic(1)">
      <span class="card-number">02</span><i class="bi bi-
box"></i>
      <h3>Variables</h3><p>Store values and learn how variables
work.</p>
      <span class="learn-link">Learn topic →</span>
    </button>
    <button class="topic-card" onclick="openTopic(2)">
      <span class="card-number">03</span><i class="bi bi-
database"></i>
      <h3>Data Types</h3><p>Choose the correct type for different
data.</p>
      <span class="learn-link">Learn topic →</span>
    </button>
    <button class="topic-card" onclick="openTopic(3)">
      <span class="card-number">04</span><i class="bi bi-
calculator"></i>

```

```

        <h3>Operators</h3><p>Perform arithmetic, comparison and
logical operations.</p>
        <span class="learn-link">Learn topic →</span>
    </button>
    <button class="topic-card" onclick="openTopic(4)">
        <span class="card-number">05</span><i class="bi bi-
signpost-split"></i>
        <h3>If / Else</h3><p>Make decisions based on a
condition.</p>
        <span class="learn-link">Learn topic →</span>
    </button>
    <button class="topic-card" onclick="openTopic(5)">
        <span class="card-number">06</span><i class="bi bi-arrow-
repeat"></i>
        <h3>Loops</h3><p>Repeat a block of code using loops.</p>
        <span class="learn-link">Learn topic →</span>
    </button>
    <button class="topic-card" onclick="openTopic(6)">
        <span class="card-number">07</span><i class="bi bi-
braces"></i>
        <h3>Functions</h3><p>Break a program into reusable
blocks.</p>
        <span class="learn-link">Learn topic →</span>
    </button>
    <button class="topic-card" onclick="openTopic(7)">
        <span class="card-number">08</span><i class="bi bi-grid-
3x3-gap"></i>
        <h3>Arrays</h3><p>Store multiple values of the same
type.</p>
        <span class="learn-link">Learn topic →</span>
    </button>
</div>
</section>

<section class="programs-section section" id="programsSection">
    <div class="section-heading">

```

```
<p class="eyebrow">6 BEGINNER PROGRAMS</p>

<h2>Practice Simple C Programs</h2>

<p>Choose a program to see what it does and how the code
works.</p>

</div>

<div class="program-list">

    <button class="program-card" onclick="openProgram(0)">

        <span>01</span><div><h3>Hello World</h3><p>Print your first
message.</p></div><b>→</b>

    </button>

    <button class="program-card" onclick="openProgram(1)">

        <span>02</span><div><h3>Addition</h3><p>Add two
numbers.</p></div><b>→</b>

    </button>

    <button class="program-card" onclick="openProgram(2)">

        <span>03</span><div><h3>Even / Odd</h3><p>Check whether a
number is even or odd.</p></div><b>→</b>

    </button>

    <button class="program-card" onclick="openProgram(3)">

        <span>04</span><div><h3>Largest Number</h3><p>Find the
largest of two numbers.</p></div><b>→</b>

    </button>

    <button class="program-card" onclick="openProgram(4)">

        <span>05</span><div><h3>Factorial</h3><p>Calculate a
number's factorial.</p></div><b>→</b>

    </button>

    <button class="program-card" onclick="openProgram(5)">

        <span>06</span><div><h3>Multiplication Table</h3><p>Print a
multiplication table using a loop.</p></div><b>→</b>

    </button>

</div>

</section>

<section class="why-section section">
```

```

    <div class="section-heading">
        <p class="eyebrow">WHY C?</p>
        <h2>Why Should You Learn C?</h2>
    </div>

    <div class="why-grid">
        <div class="why-card"><i class="bi bi-lightbulb"></i><h3>Strong
Basics</h3><p>C makes core programming concepts easier to understand.</p></div>

        <div class="why-card"><i class="bi bi-cpu"></i><h3>Fast
Language</h3><p>C is known for being fast and efficient.</p></div>

        <div class="why-card"><i class="bi bi-laptop"></i><h3>Useful
Skill</h3><p>C gives you a foundation for learning other languages.</p></div>
    </div>

</section>

</main>

<footer>
    <div class="footer-container">
        <div><h3>Learn C</h3><p>A simple learning website for beginners who
want to learn C Programming.</p></div>

        <div><h4>Quick Links</h4><button
onclick="goHome()">Home</button><button
onclick="showSection('topicsSection')">Topics</button><button
onclick="showSection('programsSection')">Programs</button></div>

        <div><h5>
            Contact
        </h5>
        <p>
            Email: 253-16-019@diu.edu.bd
        </p>
        <p>
            Learn C. Build Skills.
        </p></div>
    </div>

    <div class="copyright">© Daffodil International University • C
Learn</div>

</footer>

```

```

<!-- Topic / Program modal -->
<div class="modal" id="contentModal" onclick="closeModalOutside(event)">
  <div class="modal-box large-modal">
    <button class="close-btn" onclick="closeModal()">×</button>
    <p class="eyebrow" id="modalLabel">TOPIC</p>
    <h2 id="modalTitle">Title</h2>
    <p id="modalDescription" class="modal-description"></p>
    <div id="modalBody"></div>
  </div>
</div>

<!-- Login / Sign Up modal -->
<div class="modal" id="authModal" onclick="closeAuthOutside(event)">
  <div class="modal-box auth-box">
    <button class="close-btn" onclick="closeAuth()">×</button>
    <div class="auth-icon"><i class="bi bi-person-circle"></i></div>
    <p class="eyebrow" id="authLabel">WELCOME BACK</p>
    <h2 id="authTitle">Log In</h2>
    <p id="authMessage" class="auth-message">Use your name and password
to continue.</p>

    <form id="authForm" onsubmit="handleAuth(event)">
      <div id="nameField" class="form-group hidden">
        <label for="authName">Name</label>
        <input type="text" id="authName" placeholder="Enter your
name">
      </div>

      <div class="form-group">
        <label for="authEmail">Email</label>
        <input type="email" id="authEmail" placeholder="Enter your
email" required>

```

```
        </div>

        <div class="form-group">
            <label for="authPassword">Password</label>
            <input type="password" id="authPassword" placeholder="Enter
your password" required>
        </div>

        <button class="primary-btn full-btn" type="submit"
id="authSubmit">Log In</button>
    </form>

    <button class="switch-auth" onclick="switchAuth()">Create an
account / Log in</button>

    <p id="authStatus" class="status-text"></p>
</div>
</div>

<script src="script.js"></script>
</body>
</html>
```

Style.css

```
* {  
    margin: 0;  
    padding: 0;  
    box-sizing: border-box;  
}  
  
html {  
    scroll-behavior: smooth;  
}  
  
body {  
    font-family: Arial, sans-serif;  
    color: #20242a;  
    background: #ffffff;  
    line-height: 1.6;  
}  
  
button,  
a,  
input {  
    font: inherit;  
}  
  
button {  
    cursor: pointer;  
}  
  
.navbar {  
    width: 100%;  
    background: #376ec6;  
    position: sticky;
```

```
    top: 0;

    z-index: 1000;

    box-shadow: 0 2px 12px rgba(0, 0, 0, 0.08);
}

.logo span {
    border: 3px solid #eef0f3;
    padding: 5px;
    margin-right: 2px;
}

.nav-container {
    max-width: 1250px;
    min-height: 78px;
    margin: auto;
    padding: 10px 25px;
    display: flex;
    align-items: center;
    gap: 28px;
}

.logo {
    display: flex;
    align-items: center;
    gap: 9px;
    color: white;
    text-decoration: none;
    font-size: 25px;
    font-weight: 700;
    white-space: nowrap;
}

.logo img {
```

```
width: 42px;
height: 42px;
object-fit: contain;
border-radius: 8px;
background: white;
}

.nav-menu {
  display: flex;
  align-items: center;
  gap: 23px;
  margin-left: 12px;
  flex: 1;
}

.nav-menu > a,
.nav-button {
  color: white;
  text-decoration: none;
  border: 0;
  background: none;
  font-size: 16px;
  font-weight: 600;
  padding: 8px 2px;
}

.nav-menu > a:hover,
.nav-button:hover {
  color: #101010;
}

.disabled-link {
```

```
    opacity: 0.6;
    cursor: not-allowed;
}

.dropdown {
    position: relative;
}

.nav-button span {
    font-size: 9px;
    margin-left: 3px;
}

.dropdown-menu {
    display: none;
    position: absolute;
    top: 42px;
    left: -12px;
    width: 215px;
    padding: 7px 0;
    background: white;
    border-radius: 10px;
    box-shadow: 0 12px 30px rgba(0, 0, 0, 0.16);
    border: 1px solid #e8edf3;
}

.dropdown-menu.show {
    display: block;
}

.dropdown-menu button {
    display: block;
```

```
width: 100%;  
padding: 10px 15px;  
text-align: left;  
background: white;  
border: 0;  
color: #25282c;  
font-size: 14px;  
}  
  
.dropdown-menu button:hover {  
    background: #eef5ff;  
    color: #0d6efd;  
}  
  
.nav-actions {  
    display: flex;  
    align-items: center;  
    gap: 13px;  
}  
  
.login-btn,  
.signup-btn {  
    border: 0;  
    border-radius: 7px;  
    font-weight: 700;  
    padding: 10px 17px;  
}  
  
.login-btn {  
    background: transparent;  
    color: white;  
}
```

```
.login-btn:hover {
    background: rgba(255, 255, 255, 0.12);
}

.signup-btn {
    background: white;
    color: #1e5db0;
}

.signup-btn:hover {
    background: #e9f2ff;
    transform: translateY(-1px);
}

.hero {
    background: #e9eef3;
    background-image: linear-gradient(rgba(233, 238, 243, 0.86), rgba(233, 238, 243, 0.92)), url("C.png");
    background-repeat: no-repeat;
    background-position: 91% 50%;
    background-size: 310px;
    padding: 90px 20px;
}

.hero-container,
.section,
.footer-container {
    max-width: 1120px;
    margin: auto;
}

.hero-container {
```

```
    display: grid;
    grid-template-columns: 1.1fr 0.9fr;
    align-items: center;
    gap: 55px;
}

.eyebrow {
    color: #0d6efd;
    font-size: 13px;
    font-weight: 800;
    letter-spacing: 2px;
    margin-bottom: 8px;
}

.hero h1 {
    font-size: 52px;
    line-height: 1.15;
    margin-bottom: 18px;
}

.hero h1 span {
    color: #0d6efd;
}

.hero-text {
    max-width: 590px;
    color: #555d66;
    font-size: 17px;
    margin-bottom: 25px;
}

.hero-buttons {
```

```
    display: flex;
    gap: 12px;
    flex-wrap: wrap;
}

.primary-btn,
.outline-btn {
    border-radius: 7px;
    padding: 12px 20px;
    font-weight: 700;
    border: 1px solid #0d6efd;
    transition: 0.2s;
}

.primary-btn {
    background: #0d6efd;
    color: white;
}

.outline-btn {
    background: white;
    color: #0d6efd;
}

.primary-btn:hover,
.outline-btn:hover {
    transform: translateY(-2px);
}

.hero-note {
    margin-top: 17px;
    color: #69737d;
```

```
    font-size: 14px;
}

.code-card {
    background: #1d2126;
    color: #f5f5f5;
    border-radius: 13px;
    overflow: hidden;
    box-shadow: 0 15px 35px rgba(0, 0, 0, 0.17);
}

.code-header {
    height: 42px;
    background: #30343a;
    display: flex;
    align-items: center;
    padding: 0 15px;
    gap: 6px;
}

.code-header span {
    width: 11px;
    height: 11px;
    border-radius: 50%;
    background: #fff;
}

.code-header small {
    margin-left: auto;
    color: #c8cdd2;
}
```

```
.code-card pre {
    padding: 25px;
    overflow-x: auto;
    font-size: 14px;
    line-height: 1.8;
}

.section {
    padding: 78px 20px;
}

.section-heading {
    text-align: center;
    margin-bottom: 42px;
}

.section-heading h2 {
    font-size: 35px;
    margin-bottom: 8px;
}

.section-heading > p:last-child {
    max-width: 680px;
    margin: auto;
    color: #69737d;
}

.intro-grid {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 55px;
    align-items: center;
```

```
}

.intro-text h3 {
    font-size: 27px;
    margin-bottom: 14px;
}

.intro-text p {
    color: #606a73;
    margin-bottom: 15px;
}

.info-box {
    background: #e9eef3;
    padding: 22px 28px;
    border-radius: 13px;
}

.info-item {
    display: flex;
    gap: 17px;
    padding: 18px 0;
    border-bottom: 1px solid #c6cdd4;
}

.info-item:last-child {
    border-bottom: 0;
}

.info-item i {
    color: #0d6efd;
    font-size: 28px;
```

```
}

.info-item h4 {
    margin-bottom: 3px;
}

.info-item p {
    color: #69737d;
    margin: 0;
}

.topics-section {
    max-width: none;
    background: #f5f8fc;
    padding-left: max(20px, calc((100% - 1120px) / 2));
    padding-right: max(20px, calc((100% - 1120px) / 2));
}

.topics-section .section-heading,
.programs-section .section-heading,
.why-section .section-heading {
    max-width: 1120px;
    margin-left: auto;
    margin-right: auto;
}

.card-grid {
    max-width: 1120px;
    margin: auto;
    display: grid;
    grid-template-columns: repeat(4, 1fr);
    gap: 18px;
}
```

```
}

.topic-card {
    position: relative;
    text-align: left;
    background: white;
    border: 1px solid #e3e8ee;
    border-radius: 12px;
    padding: 24px 20px;
    min-height: 245px;
    transition: 0.2s;
}

.topic-card:hover {
    transform: translateY(-5px);
    border-color: #a9caff;
    box-shadow: 0 12px 25px rgba(31, 90, 170, 0.11);
}

.topic-card i {
    display: block;
    color: #0d6efd;
    font-size: 34px;
    margin: 22px 0 13px;
}

.card-number {
    position: absolute;
    top: 17px;
    right: 17px;
    color: #aab5c0;
    font-weight: 800;
}
```

```
        font-size: 13px;
    }

    .topic-card h3 {
        font-size: 19px;
        margin-bottom: 7px;
    }

    .topic-card p {
        color: #69737d;
        font-size: 14px;
    }

    .learn-link {
        color: #0d6efd;
        font-size: 14px;
        font-weight: 700;
    }

    .programs-section {
        background: white;
    }

    .program-list {
        max-width: 900px;
        margin: auto;
        display: grid;
        grid-template-columns: 1fr 1fr;
        gap: 15px;
    }

    .program-card {
```

```
display: grid;
grid-template-columns: 48px 1fr 20px;
align-items: center;
gap: 14px;
text-align: left;
border: 1px solid #ele6eb;
background: white;
border-radius: 11px;
padding: 17px;
transition: 0.2s;
}

.program-card:hover {
border-color: #9fc4ff;
box-shadow: 0 8px 20px rgba(0, 0, 0, 0.07);
transform: translateY(-2px);
}

.program-card > span {
width: 42px;
height: 42px;
display: grid;
place-items: center;
border-radius: 9px;
background: #eef5ff;
color: #0d6efd;
font-weight: 800;
}

.program-card h3 {
font-size: 17px;
margin-bottom: 2px;
```

```
}

.program-card p {
    color: #707a83;
    font-size: 13px;
    margin: 0;
}

.program-card b {
    color: #0d6efd;
    font-size: 20px;
}

.why-section {
    background: #f5f8fc;
}

.why-grid {
    display: grid;
    grid-template-columns: repeat(3, 1fr);
    gap: 18px;
}

.why-card {
    text-align: center;
    background: white;
    padding: 30px 24px;
    border: 1px solid #e5e9ee;
    border-radius: 12px;
}

.why-card i {
```

```
    color: #0d6efd;
    font-size: 38px;
}

.why-card h3 {
    margin: 12px 0 8px;
}

.why-card p {
    color: #69737d;
}

footer {
    background: #212529;
    color: white;
    padding: 50px 20px 18px;
}

.footer-container {
    display: grid;
    grid-template-columns: 1.5fr 1fr 1fr;
    gap: 40px;
}

.footer-container h3,
.footer-container h4 {
    margin-bottom: 12px;
}

.footer-container p,
.footer-container button {
    color: #c9ced3;
```

```
}

.footer-container p {
    margin-bottom: 5px;
}

.footer-container button {
    display: block;
    border: 0;
    background: none;
    padding: 2px 0;
    margin-bottom: 5px;
}

.footer-container button:hover {
    color: white;
}

.copyright {
    max-width: 1120px;
    margin: 35px auto 0;
    padding-top: 18px;
    border-top: 1px solid #4d5257;
    text-align: center;
    color: #bfc5ca;
    font-size: 13px;
}

.modal {
    display: none;
    position: fixed;
    inset: 0;
```

```
    z-index: 2000;

    background: rgba(10, 20, 30, 0.58);

    padding: 25px;

    overflow-y: auto;
}

.modal.show {

    display: flex;

    align-items: center;

    justify-content: center;
}

.modal-box {

    position: relative;

    width: 100%;

    max-width: 760px;

    background: white;

    border-radius: 15px;

    padding: 35px;

    box-shadow: 0 20px 50px rgba(0, 0, 0, 0.2);
}

.auth-box {

    max-width: 450px;
}

.close-btn {

    position: absolute;

    top: 10px;

    right: 14px;

    border: 0;

    background: none;
```

```
    color: #69737d;
    font-size: 30px;
    line-height: 1;
}

.modal-box h2 {
    font-size: 30px;
    margin-bottom: 8px;
}

.modal-description {
    color: #66717b;
    margin-bottom: 25px;
}

.detail-grid {
    display: grid;
    grid-template-columns: 0.9fr 1.1fr;
    gap: 25px;
}

.detail-grid h3,
.program-detail h3 {
    font-size: 18px;
    margin-bottom: 10px;
}

.point-list {
    padding-left: 20px;
    color: #5e6871;
}
```

```
.point-list li {  
    margin-bottom: 9px;  
}  
  
.code-example {  
    background: #1d2126;  
    color: #f4f5f6;  
    padding: 18px;  
    border-radius: 9px;  
    overflow-x: auto;  
    line-height: 1.7;  
    font-size: 13px;  
}  
  
.output-box {  
    background: #eef5ff;  
    color: #195ba8;  
    padding: 14px;  
    border-radius: 8px;  
    font-weight: 700;  
    margin-top: 8px;  
}  
  
.auth-icon {  
    color: #0d6efd;  
    font-size: 42px;  
    margin-bottom: 5px;  
}  
  
.auth-message {  
    color: #69737d;  
    font-size: 14px;
```

```
        margin-bottom: 20px;
    }

    .form-group {
        margin-bottom: 15px;
    }

    .form-group label {
        display: block;
        font-weight: 700;
        font-size: 14px;
        margin-bottom: 6px;
    }

    .form-group input {
        width: 100%;
        border: 1px solid #d5dce3;
        border-radius: 7px;
        padding: 11px 12px;
        outline: none;
    }

    .form-group input:focus {
        border-color: #0d6efd;
    }

    .hidden {
        display: none;
    }

    .full-btn {
        width: 100%;
    }
```

```
        margin-top: 5px;
    }

    .switch-auth {
        width: 100%;
        border: 0;
        background: none;
        color: #0d6efd;
        margin-top: 15px;
        font-size: 14px;
    }

    .status-text {
        text-align: center;
        color: #198754;
        font-size: 14px;
        min-height: 20px;
        margin-top: 10px;
    }

    @media (max-width: 1050px) {
        .nav-container {
            gap: 15px;
        }

        .nav-menu {
            gap: 15px;
        }

        .card-grid {
            grid-template-columns: repeat(2, 1fr);
        }
    }
```

```
}

@media (max-width: 820px) {
  .nav-container {
    flex-wrap: wrap;
    justify-content: center;
    padding: 12px 18px;
  }

  .logo {
    margin-right: auto;
  }

  .nav-actions {
    margin-left: auto;
  }

  .nav-menu {
    order: 3;
    flex-basis: 100%;
    justify-content: center;
    margin-left: 0;
    padding-top: 4px;
  }

  .hero-container,
  .intro-grid {
    grid-template-columns: 1fr;
  }

  .hero {
    text-align: center;
  }
}
```

```
        background-position: 50% 88%;
        background-size: 220px;
    }

    .hero-text {
        margin-left: auto;
        margin-right: auto;
    }

    .hero-buttons {
        justify-content: center;
    }

    .code-card {
        max-width: 600px;
        width: 100%;
        margin: auto;
    }

    .why-grid {
        grid-template-columns: 1fr;
    }

    .footer-container {
        grid-template-columns: 1fr 1fr;
    }
}

@media (max-width: 600px) {
    .nav-container {
        padding: 10px 14px;
    }
}
```

```
.logo {  
    font-size: 21px;  
}  
  
.logo img {  
    width: 36px;  
    height: 36px;  
}  
  
.nav-menu {  
    gap: 10px;  
    flex-wrap: wrap;  
}  
  
.nav-menu > a,  
.nav-button {  
    font-size: 13px;  
}  
  
.dropdown-menu {  
    left: 50%;  
    transform: translateX(-50%);  
    top: 36px;  
}  
  
.nav-actions {  
    gap: 5px;  
}  
  
.login-btn,  
.signup-btn {
```

```
padding: 8px 11px;
font-size: 13px;
}

.hero {
padding: 60px 16px;
}

.hero h1 {
font-size: 34px;
}

.hero-text {
font-size: 15px;
}

.section {
padding: 60px 16px;
}

.section-heading h2 {
font-size: 29px;
}

.card-grid,
.program-list,
.footer-container {
grid-template-columns: 1fr;
}

.topic-card {
min-height: 210px;
```

```
}

.modal {
  padding: 12px;
}

.modal-box {
  padding: 28px 20px 22px;
}

.detail-grid {
  grid-template-columns: 1fr;
}
}

@media (max-width: 380px) {
  .hero h1 {
    font-size: 29px;
  }

  .nav-actions {
    width: 100%;
    justify-content: flex-end;
  }

  .nav-menu {
    row-gap: 5px;
  }
}
```

script.js

```
let currentAuth = "login";
const topics = [
  {
    title: "C Introduction",
    description: "C is a general-purpose programming language. A C program normally starts execution from main().",
    syntax: "int main()\n{\n    // statements\n    return 0;\n}",
    points: [
      "C programs use statements inside functions.",
      "#include <stdio.h> gives access to printf() and scanf().",
      "main() is the starting point of a normal C program."
    ]
  },
  {
    title: "Variables",
    description: "A variable is a named place in memory where a value can be stored.",
    syntax: "int age = 20;\nfloat price = 10.5;",
    points: [
      "The variable type is written before the variable name.",
      "The value can be changed during program execution.",
      "Example: int marks = 80;"
    ]
  },
  {
    title: "Data Types",
    description: "Data types tell C what kind of value a variable will store.",
    syntax: "int age = 20;\nfloat cgpa = 3.50;\nchar grade = 'A';",
```

```
    points: [
        "int is commonly used for whole numbers.",
        "float is used for decimal numbers.",
        "char stores a single character."
    ]
},
{
    title: "Operators",
    description: "Operators are symbols used to perform operations on
values.",
    syntax: "sum = a + b;\nremainder = a % b;\nif (a > b)",
    points: [
        "Arithmetic: +, -, *, /, %",
        "Relational: >, <, >=, <=, ==, !=",
        "Assignment: ="
    ]
},
{
    title: "If / Else",
    description: "if / else lets a program make a decision depending on a
condition.",
    syntax: "if (number > 0)\n{\n    printf(\"Positive\");\n}\nelse\n{\n
printf(\"Not positive\");\n}",
    points: [
        "The if block runs when the condition is true.",
        "The else block runs when the condition is false.",
        "Conditions commonly use relational operators."
    ]
},
{
    title: "Loops",
    description: "Loops repeat a block of statements. for, while and do-
while are common C loops.",
    syntax: "for (int i = 1; i <= 5; i++)\n{\n    printf(\"%d \", i);\n}",
```

```
        points: [
            "Use a loop when the same task must be repeated.",
            "for is useful when the number of repetitions is known.",
            "The loop condition controls when repetition stops."
        ]
    },
    {
        title: "Functions",
        description: "A function is a block of code designed to perform a
particular task.",
        syntax: "void hello()\n{\n    printf(\"Hello\");\n}\n\nhello();",
        points: [
            "Functions make programs easier to organize.",
            "A function can be called when its task is needed.",
            "Functions can return a value or return void."
        ]
    },
    {
        title: "Arrays",
        description: "An array stores multiple values of the same data type
under one name.",
        syntax: "int numbers[5] = {10, 20, 30, 40, 50};\nprintf(\"%d\",
numbers[0]);",
        points: [
            "Array indexing starts from 0.",
            "numbers[0] is the first element.",
            "The array size is written inside square brackets."
        ]
    }
];

const programs = [
    {
```

```
        title: "Hello World",
        description: "Print a simple message on the screen.",
        code: '#include <stdio.h>\n\nint main()\n{\n    printf("Hello\nWorld!");\n    return 0;\n}',
        output: "Hello World!"
    },
    {
        title: "Addition",
        description: "Take two integers and display their sum.",
        code: '#include <stdio.h>\n\nint main()\n{\n    int a, b;\n    scanf("%d %d", &a, &b);\n    printf("%d", a + b);\n    return 0;\n}',
        output: "For input 5 3 → 8"
    },
    {
        title: "Even / Odd",
        description: "Use if / else and the modulus operator to check a\nnumber.",
        code: '#include <stdio.h>\n\nint main()\n{\n    int n;\n    scanf("%d", &n);\n    if (n % 2 == 0)\n        printf("Even");\n    else\n        printf("Odd");\n    return 0;\n}',
        output: "For input 8 → Even"
    },
    {
        title: "Largest Number",
        description: "Compare two numbers and print the larger one.",
        code: '#include <stdio.h>\n\nint main()\n{\n    int a, b;\n    scanf("%d %d", &a, &b);\n    if (a > b)\n        printf("%d", a);\n    else\n        printf("%d", b);\n    return 0;\n}',
        output: "For input 12 9 → 12"
    },
    {
        title: "Factorial",
        description: "Use a for loop to calculate the factorial of a number.",
        code: '#include <stdio.h>\n\nint main()\n{\n    int n, i, fact =\n1;\n    scanf("%d", &n);\n    for (i = 1; i <= n; i++)\n        fact =\n        fact * i;\n    printf("%d", fact);\n    return 0;\n}',
```

```
        output: "For input 5 → 120"
    },
    {
        title: "Multiplication Table",
        description: "Use a for loop to print a number's multiplication
table.",
        code: '#include <stdio.h>\n\nint main()\n{\n    int n, i;\n    scanf("%d", &n);\n    for (i = 1; i <= 10; i++)\n        printf("%d x %d = %d\\n", n, i, n * i);\n    return 0;\n}',
        output: "For input 5 → 5 x 1 = 5 ... 5 x 10 = 50"
    }
];

function toggleMenu(menuId) {
    let menu = document.getElementById(menuId);
    let otherMenuId = menuId === "topicsMenu" ? "programsMenu" : "topicsMenu";
    let otherMenu = document.getElementById(otherMenuId);

    otherMenu.classList.remove("show");
    menu.classList.toggle("show");
}

function closeMenus() {
    document.getElementById("topicsMenu").classList.remove("show");
    document.getElementById("programsMenu").classList.remove("show");
}

function showSection(sectionId) {
    closeMenus();
    let section = document.getElementById(sectionId);

    if (section) {
        section.scrollIntoView({ behavior: "smooth" });
    }
}
```

```
    }  
}  
  
function goHome() {  
    closeMenus();  
    window.scrollTo({ top: 0, behavior: "smooth" });  
}  
  
function openTopic(index) {  
    closeMenus();  
    let topic = topics[index];  
  
    if (topic) {  
        document.getElementById("modalLabel").textContent = "TOPIC " + (index +  
1);  
        document.getElementById("modalTitle").textContent = topic.title;  
        document.getElementById("modalDescription").textContent =  
topic.description;  
  
        let points = "";  
        for (let i = 0; i < topic.points.length; i++) {  
            points += "<li>" + topic.points[i] + "</li>";  
        }  
  
        document.getElementById("modalBody").innerHTML =  
            '<div class="detail-grid">' +  
                '<div><h3>Key Points</h3><ul class="point-list">' + points +  
'</ul></div>' +  
                '<div><h3>Simple Example</h3><pre class="code-example"><code>' +  
                escapeHtml(topic.syntax) +  
                '</code></pre></div>' +  
                '</div>';  
    }  
}
```

```
        document.getElementById("contentModal").classList.add("show");
    }
}

function openProgram(index) {
    closeMenus();
    let program = programs[index];

    if (program) {
        document.getElementById("modalLabel").textContent = "PROGRAM " + (index
+ 1);
        document.getElementById("modalTitle").textContent = program.title;
        document.getElementById("modalDescription").textContent =
program.description;

        document.getElementById("modalBody").innerHTML =
            '<div class="program-detail">' +
                '<h3>C Code</h3>' +
                '<pre class="code-example"><code>' +
escapeHtml(program.code.replace(/\n/g, "\n")) + '</code></pre>' +
                '<h3>Expected Output</h3>' +
                '<div class="output-box">' + program.output + '</div>' +
            '</div>';

        document.getElementById("contentModal").classList.add("show");
    }
}

function escapeHtml(text) {
    return text
        .replace(/&/g, "&amp;")
        .replace(/</g, "&lt;")
        .replace(/>/g, "&gt;");
}
```

```
}

function closeModal() {
    document.getElementById("contentModal").classList.remove("show");
}

function closeModalOutside(event) {
    if (event.target.id === "contentModal") {
        closeModal();
    }
}

function openAuth(type) {
    closeMenus();
    currentAuth = type;
    updateAuthForm();
    document.getElementById("authModal").classList.add("show");
}

function switchAuth() {
    if (currentAuth === "login") {
        currentAuth = "signup";
    } else {
        currentAuth = "login";
    }

    updateAuthForm();
}

function updateAuthForm() {
    let isLogin = currentAuth === "login";
    let nameField = document.getElementById("nameField");
```

```
document.getElementById("authLabel").textContent = isLogin ? "WELCOME BACK"
: "CREATE ACCOUNT";

document.getElementById("authTitle").textContent = isLogin ? "Log In" :
"Sign Up";

document.getElementById("authMessage").textContent =
    isLogin ? "Use your saved email and password to continue." : "Create a
simple account for this browser.";

document.getElementById("authSubmit").textContent = isLogin ? "Log In" :
"Sign Up";

document.querySelector(".switch-auth").textContent =
    isLogin ? "Need an account? Sign Up" : "Already have an account? Log
In";

if (isLogin) {
    nameField.classList.add("hidden");
    document.getElementById("authName").required = false;
} else {
    nameField.classList.remove("hidden");
    document.getElementById("authName").required = true;
}

document.getElementById("authStatus").textContent = "";
}

function handleAuth(event) {
    event.preventDefault();

    let email = document.getElementById("authEmail").value.trim();
    let password = document.getElementById("authPassword").value;
    let name = document.getElementById("authName").value.trim();

    if (currentAuth === "signup") {
        if (name === "") {
```

```
        document.getElementById("authStatus").textContent = "Please enter  
your name.";
        return;
    }

    localStorage.setItem("cLearnUser", JSON.stringify({
        name: name,
        email: email,
        password: password
    }));

    document.getElementById("authStatus").textContent = "Sign Up  
successful! You can now log in.";
    currentAuth = "login";
    updateAuthForm();
    document.getElementById("authEmail").value = email;
    return;
}

let savedUser = localStorage.getItem("cLearnUser");

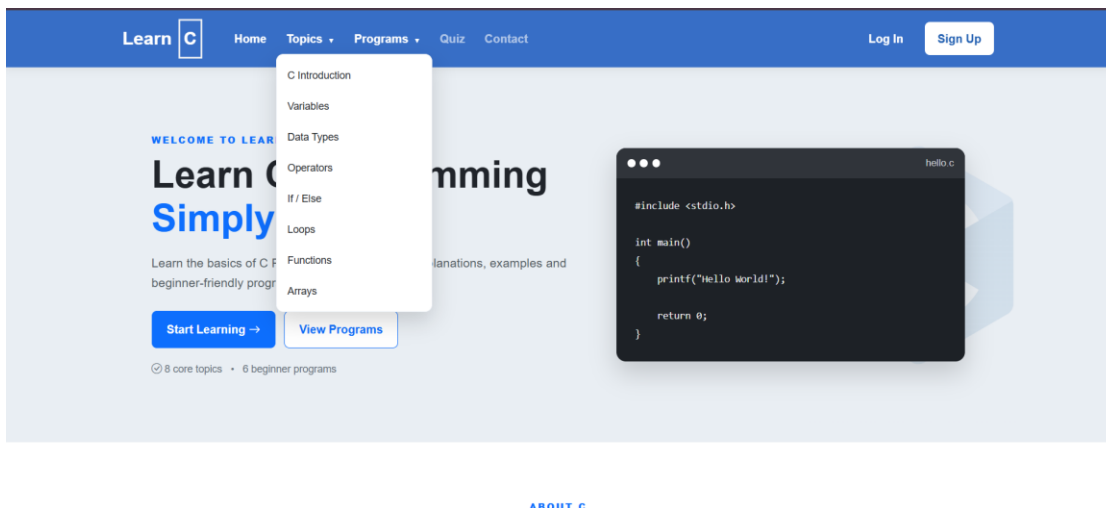
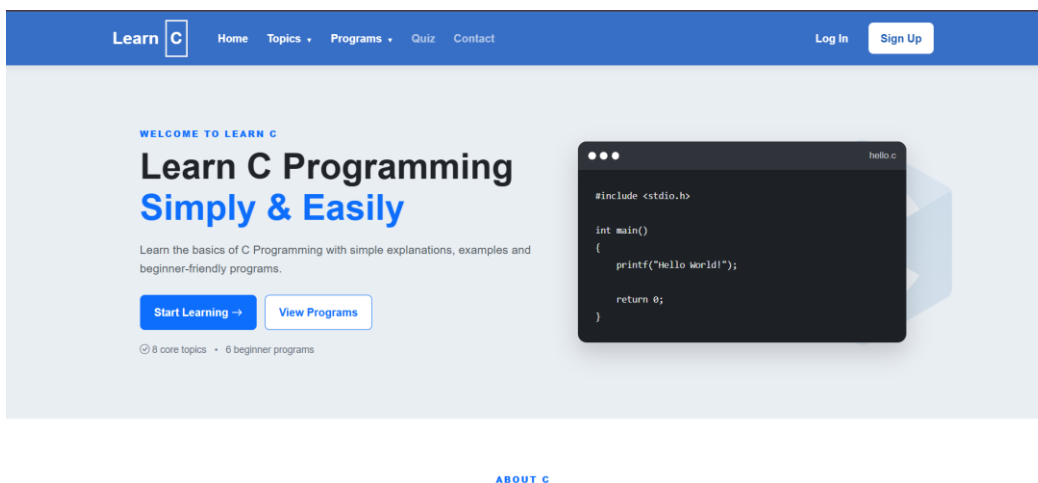
if (savedUser === null) {
    document.getElementById("authStatus").textContent = "No account found.  
Please Sign Up first.";
    return;
}

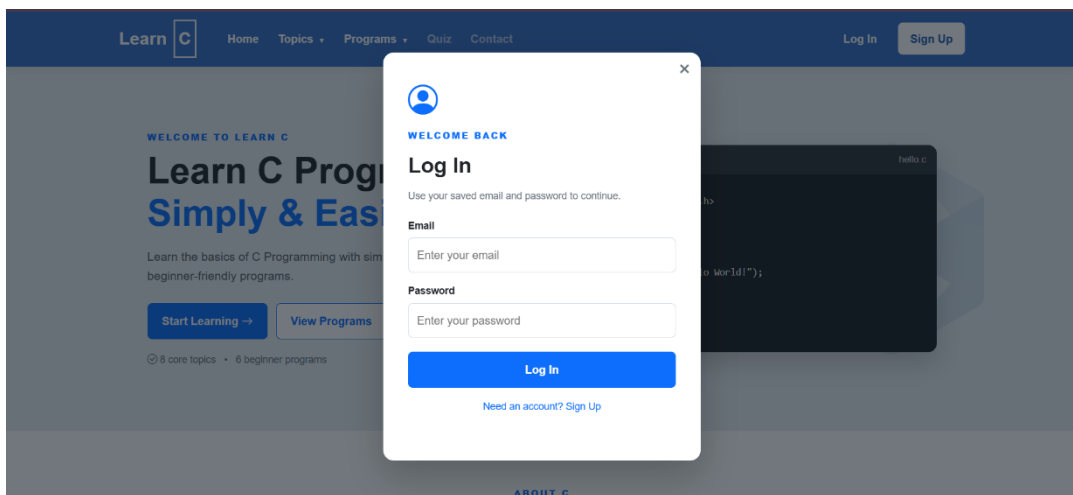
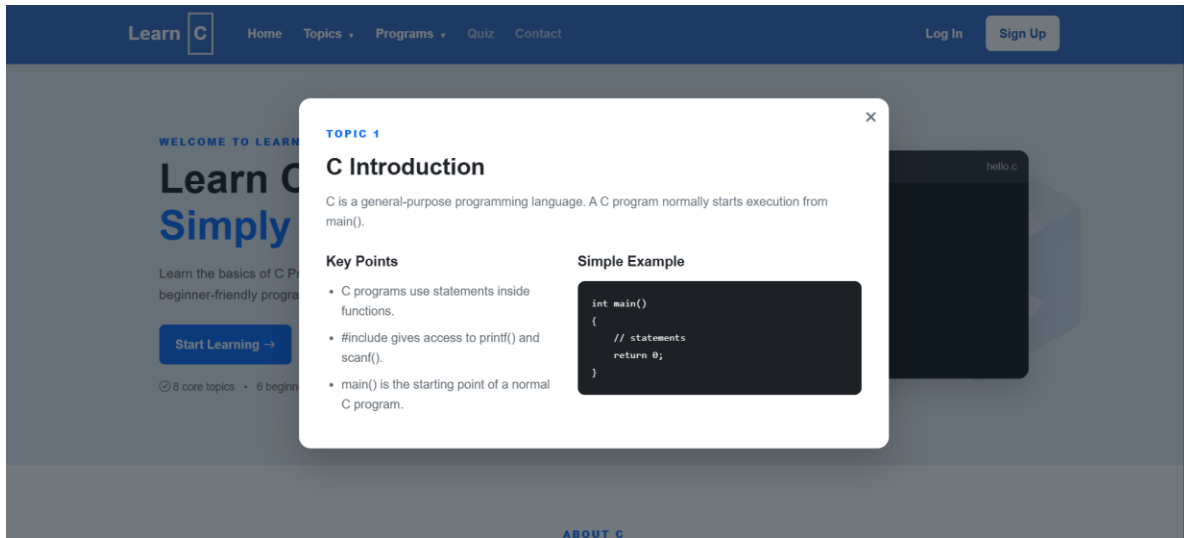
let user = JSON.parse(savedUser);

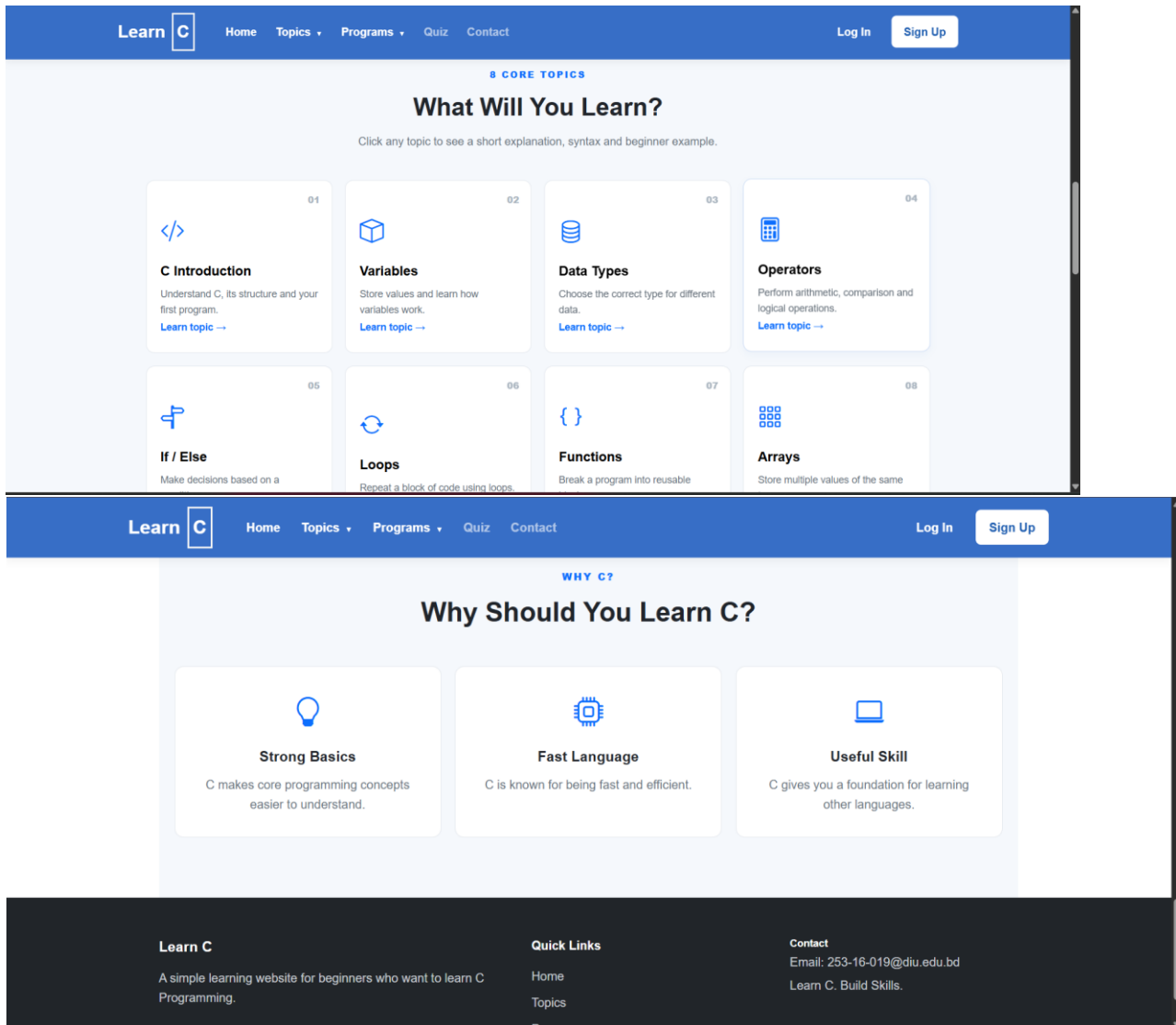
if (email === user.email && password === user.password) {
    document.getElementById("authStatus").textContent = "Login successful!  
Welcome, " + user.name + ".";
} else {
```

```
        document.getElementById("authStatus").textContent = "Email or password  
is incorrect.";  
    }  
}  
  
function closeAuth() {  
    document.getElementById("authModal").classList.remove("show");  
    document.getElementById("authForm").reset();  
}  
  
function closeAuthOutside(event) {  
    if (event.target.id === "authModal") {  
        closeAuth();  
    }  
}  
  
document.addEventListener("click", function(event) {  
    if (!event.target.closest(".dropdown")) {  
        closeMenus();  
    }  
});  
  
document.addEventListener("keydown", function(event) {  
    if (event.key === "Escape") {  
        closeModal();  
        closeAuth();  
    }  
});
```

11. Sample Output







12. Conclusion

LearnC is a simple and beginner-friendly C programming learning website.

The project combines HTML, CSS, Bootstrap, and basic JavaScript to create an interactive and responsive website.

This project helped me improve my knowledge of web development and understand how different web technologies work together.

13. References

1. **MDN Web Docs** — HTML, CSS and JavaScript documentation.
<https://developer.mozilla.org/>

2. **W3Schools** — HTML, CSS and JavaScript tutorials.
<https://www.w3schools.com/>
3. **Bootstrap Icons** — Icons used in the website.
<https://icons.getbootstrap.com/>
4. **GeeksforGeeks** — C Programming concepts and examples.
<https://www.geeksforgeeks.org/c/>
5. **C Programming Language Reference** — Basic C programming syntax and concepts.
<https://en.cppreference.com/w/c>

END

Thank You